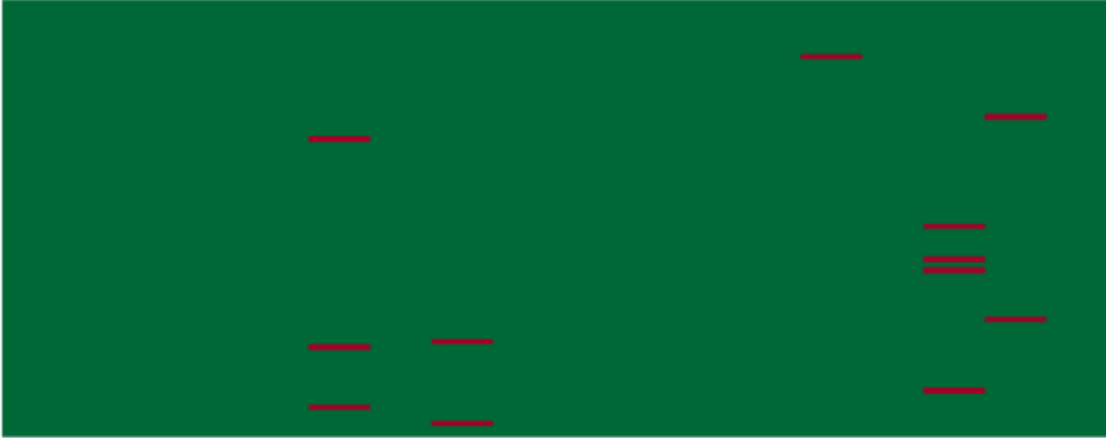


Fig 1: Missing Values — Before vs After

Missing Values Analysis

BEFORE Cleaning — Missing Values (red=missing)



AFTER Cleaning — Missing Values (all green)



Row ID
Order ID
Order Date
Ship Date
Ship Mode
Customer Name
Segment
City
State
Postal Code
Category
Sub-Category
Product Name
Sales (₹)
Quantity
Discount
Profit (₹)
Order Priority

Row ID
Order ID
Order Date
Ship Date
Ship Mode
Customer Name
Segment
City
State
Postal Code
Category
Sub-Category
Product Name
Sales (₹)
Quantity
Discount
Profit (₹)
Order Priority
Delivery Days
Order Month
Order Year
Profit Margin (%)

Fig 2: Sales & Profit by Category

Sales & Profit by Category (in ₹)

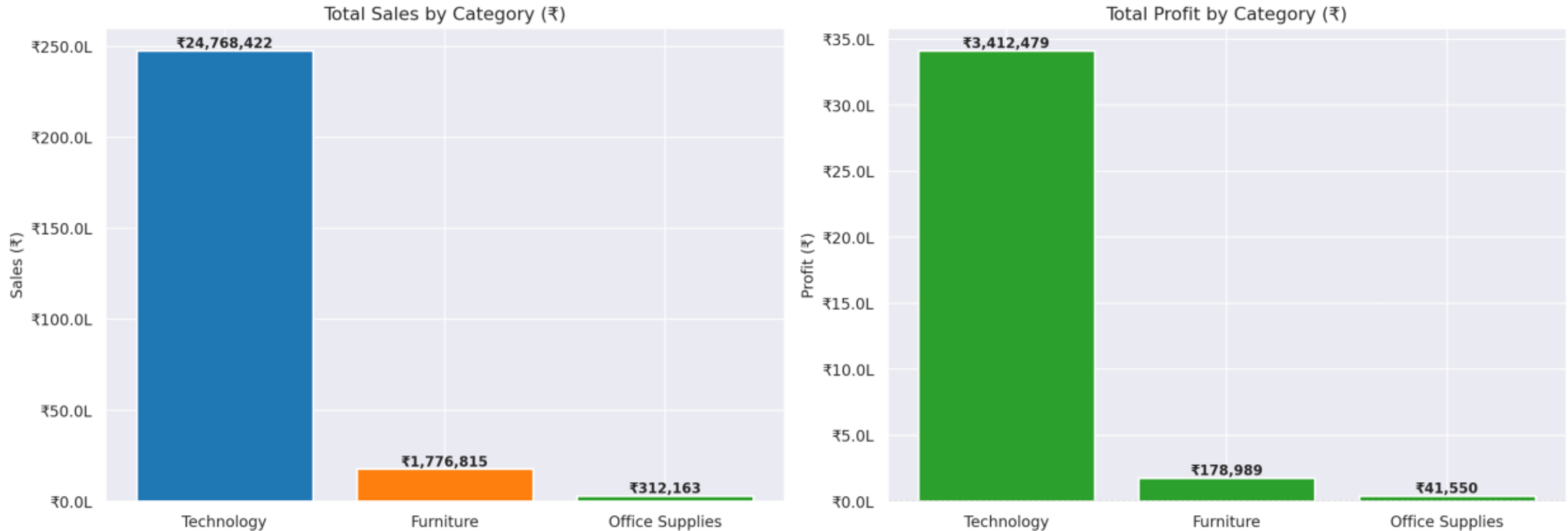


Fig 3: Monthly Sales Trend 2022-2023

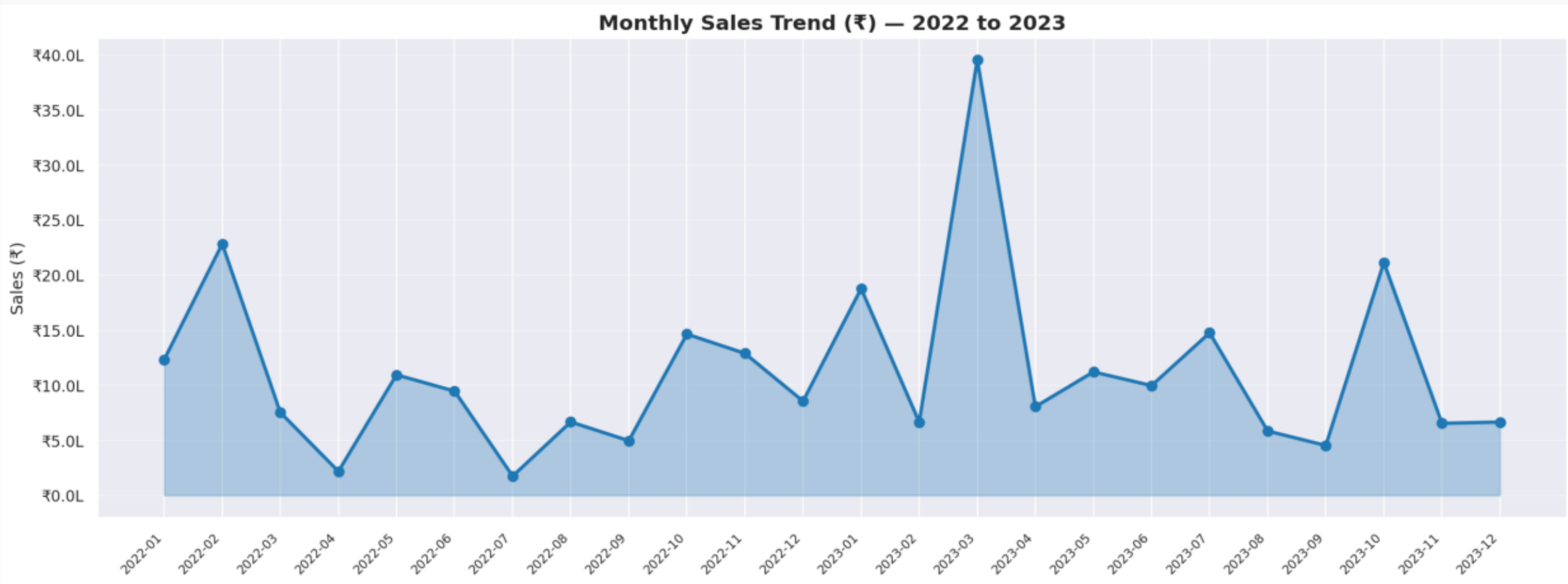


Fig 4: Segment Analysis

Segment Analysis (₹)

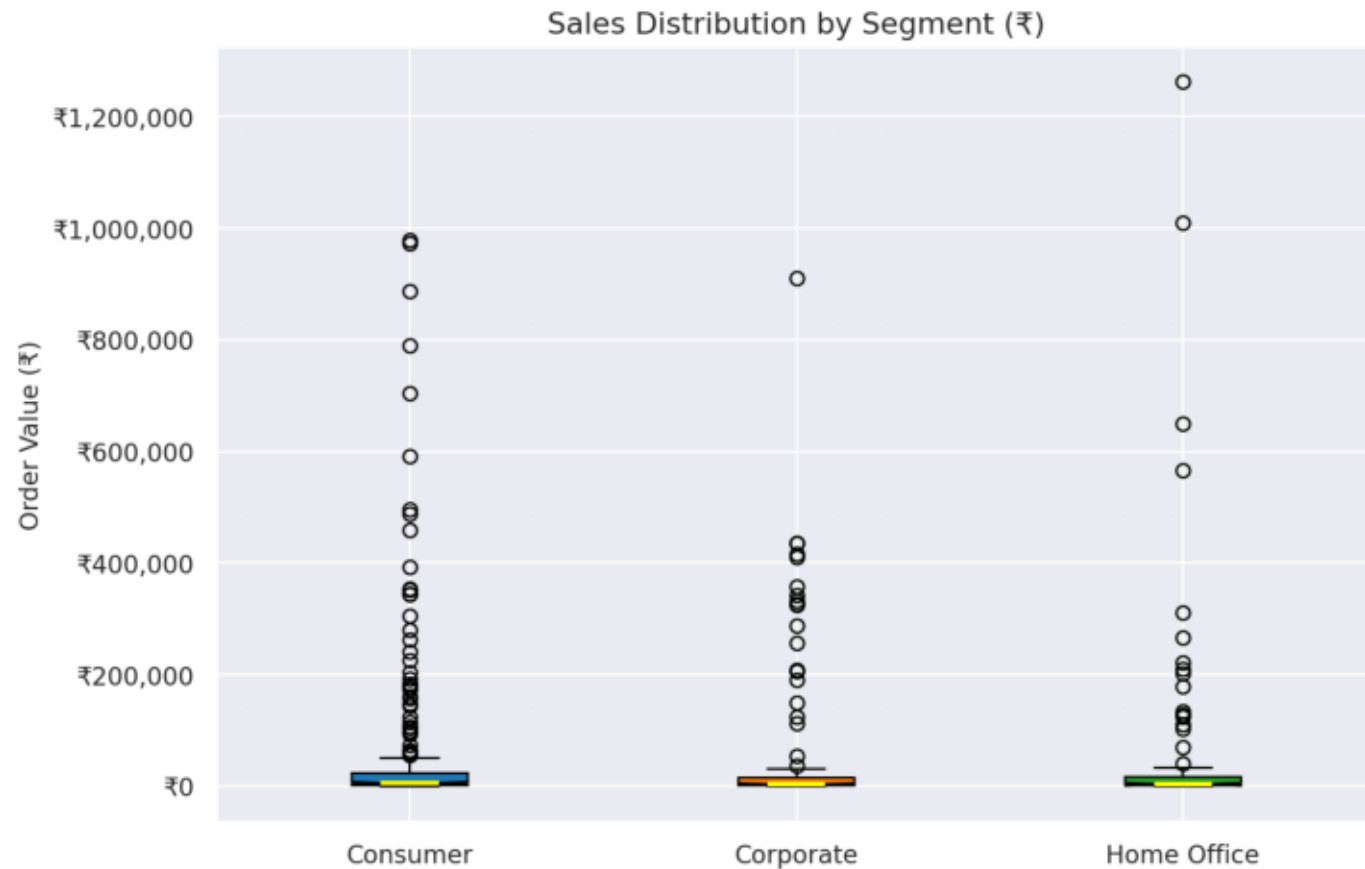
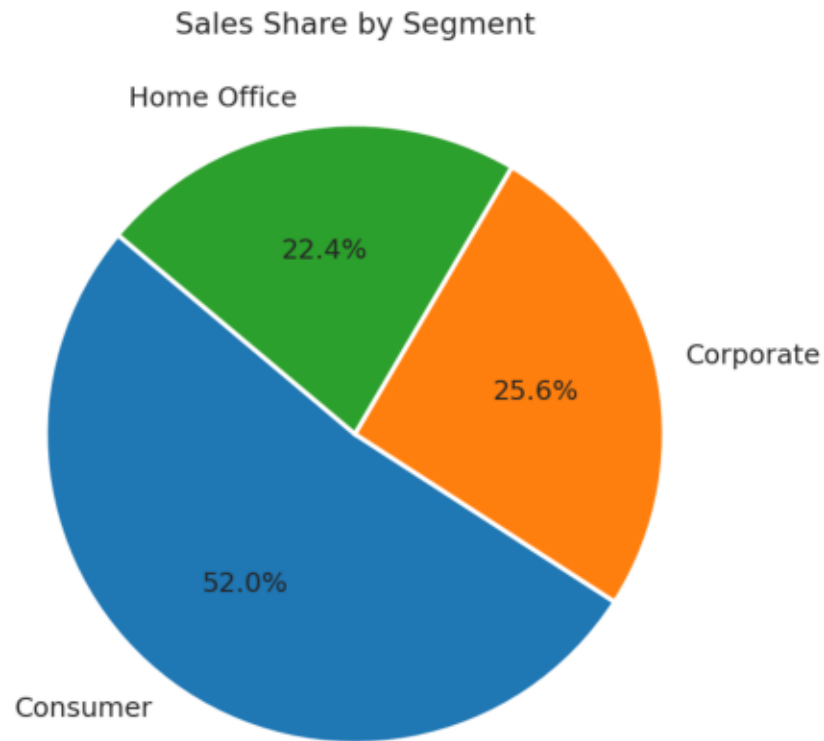
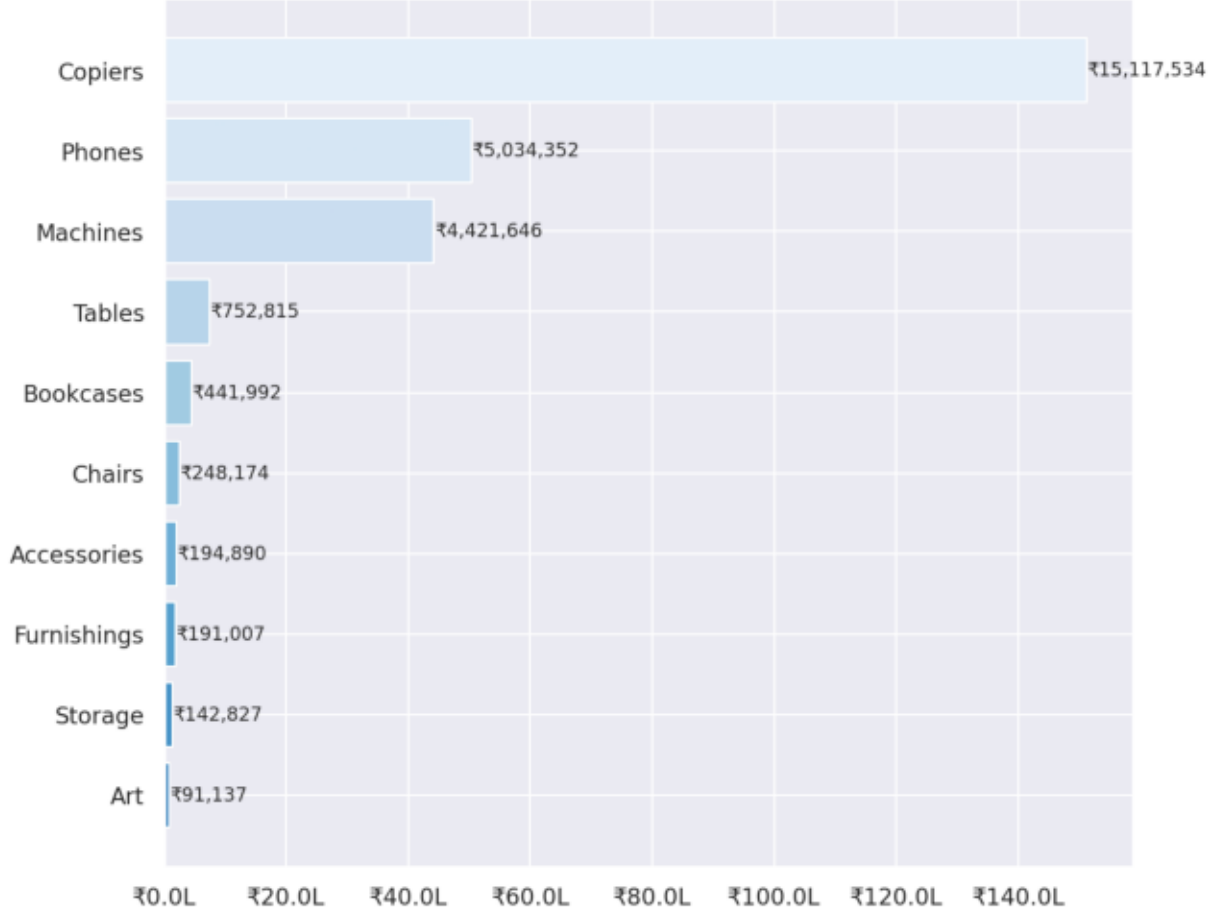


Fig 5: Sub-Category Deep Dive

Sub-Category Deep Dive (₹)

Top 10 Sub-Categories by Sales



Top 10 Sub-Categories by Profit

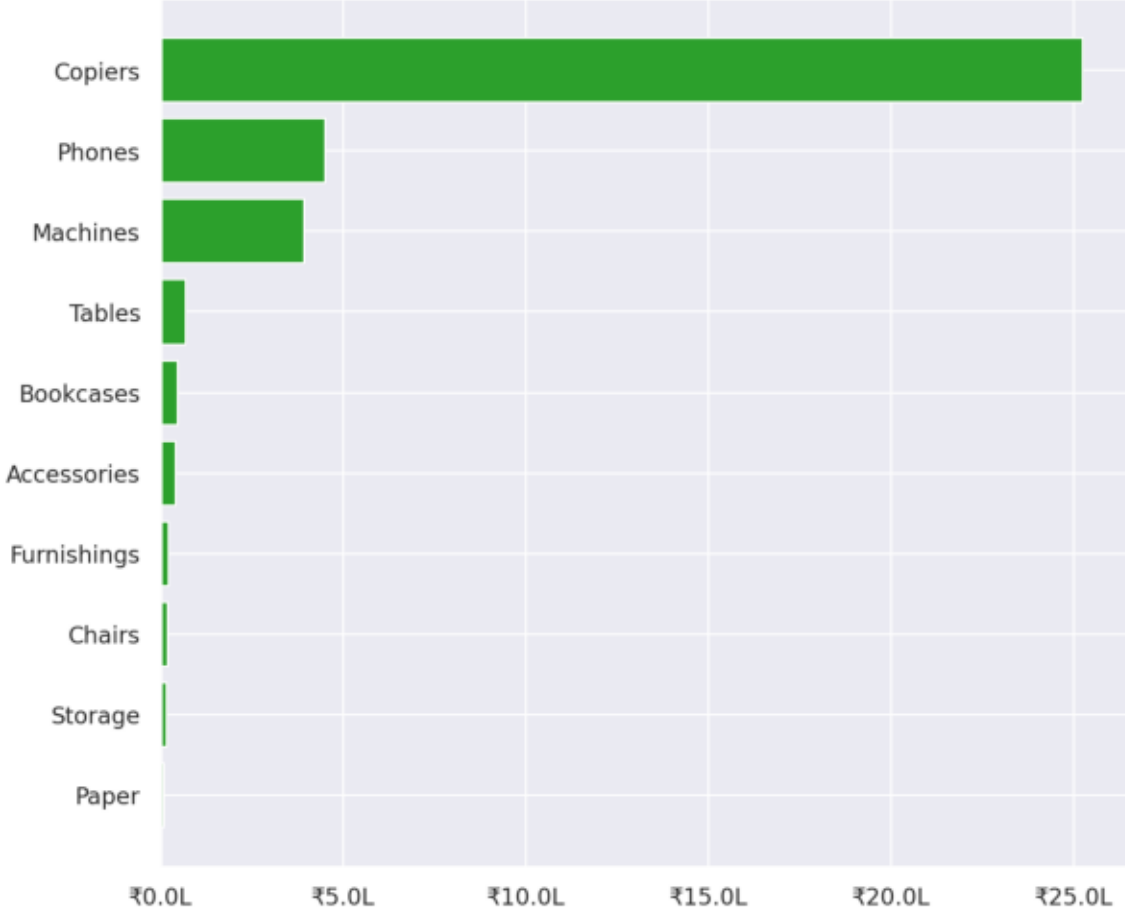


Fig 6: Sales vs Profit Scatter

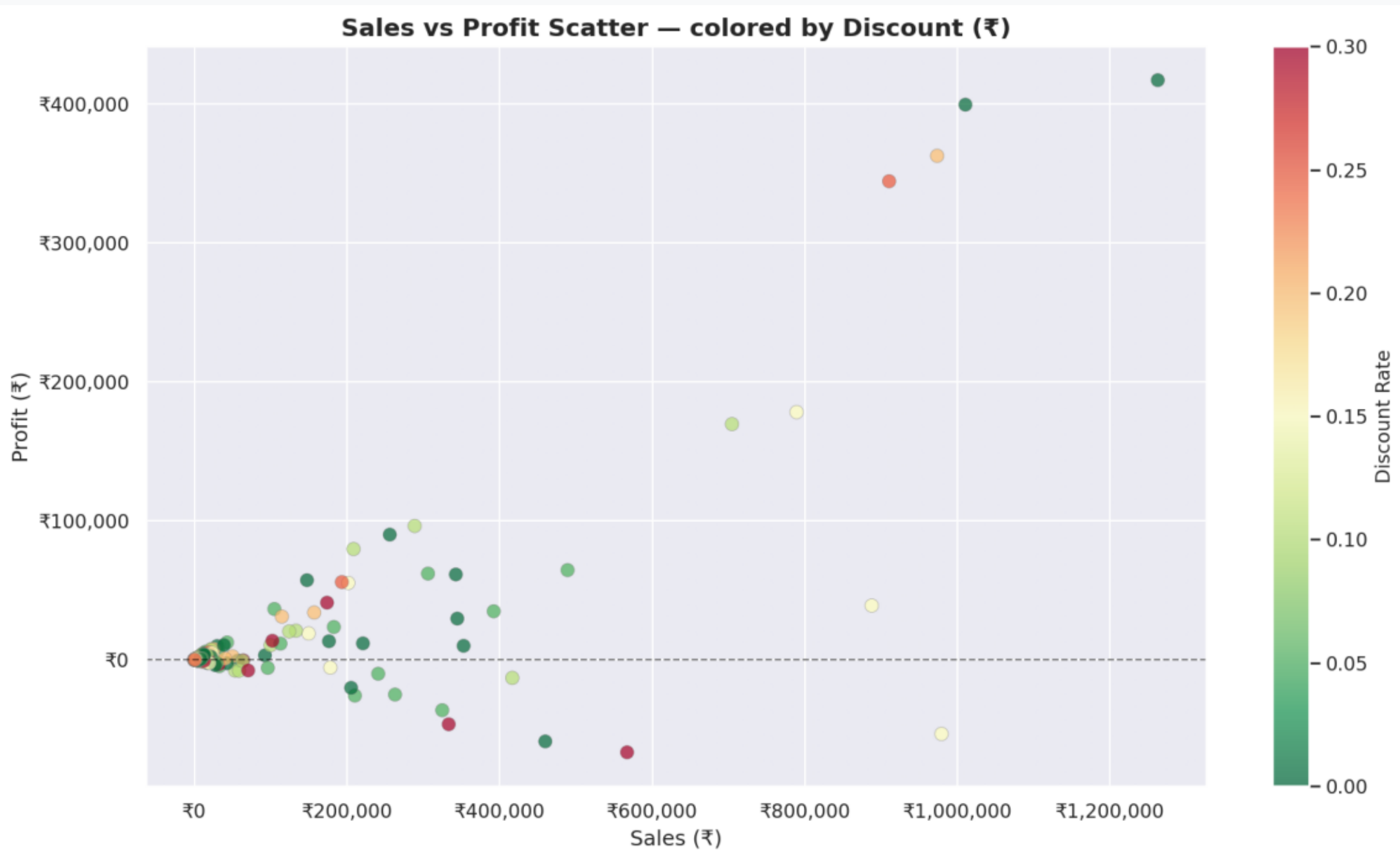
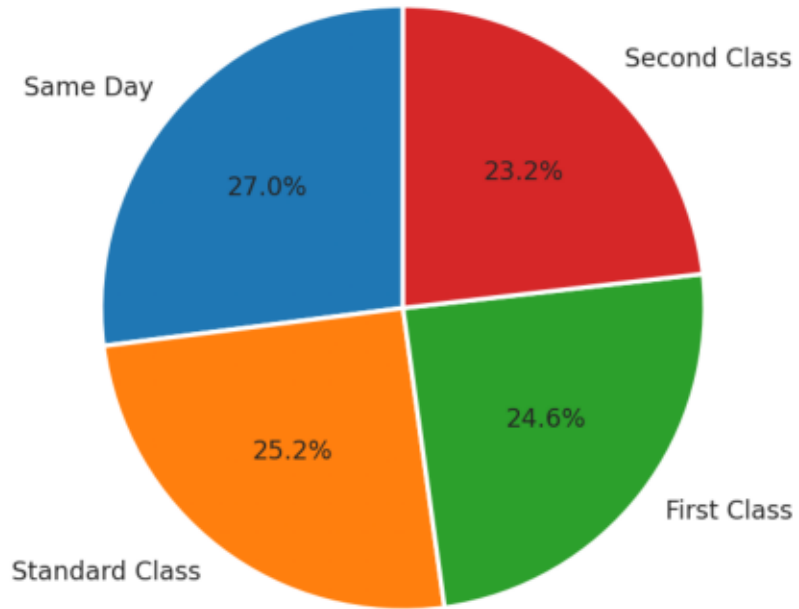


Fig 7: Shipping Mode & Delivery Days

Shipping Analysis

Orders by Ship Mode



Delivery Days Distribution by Ship Mode

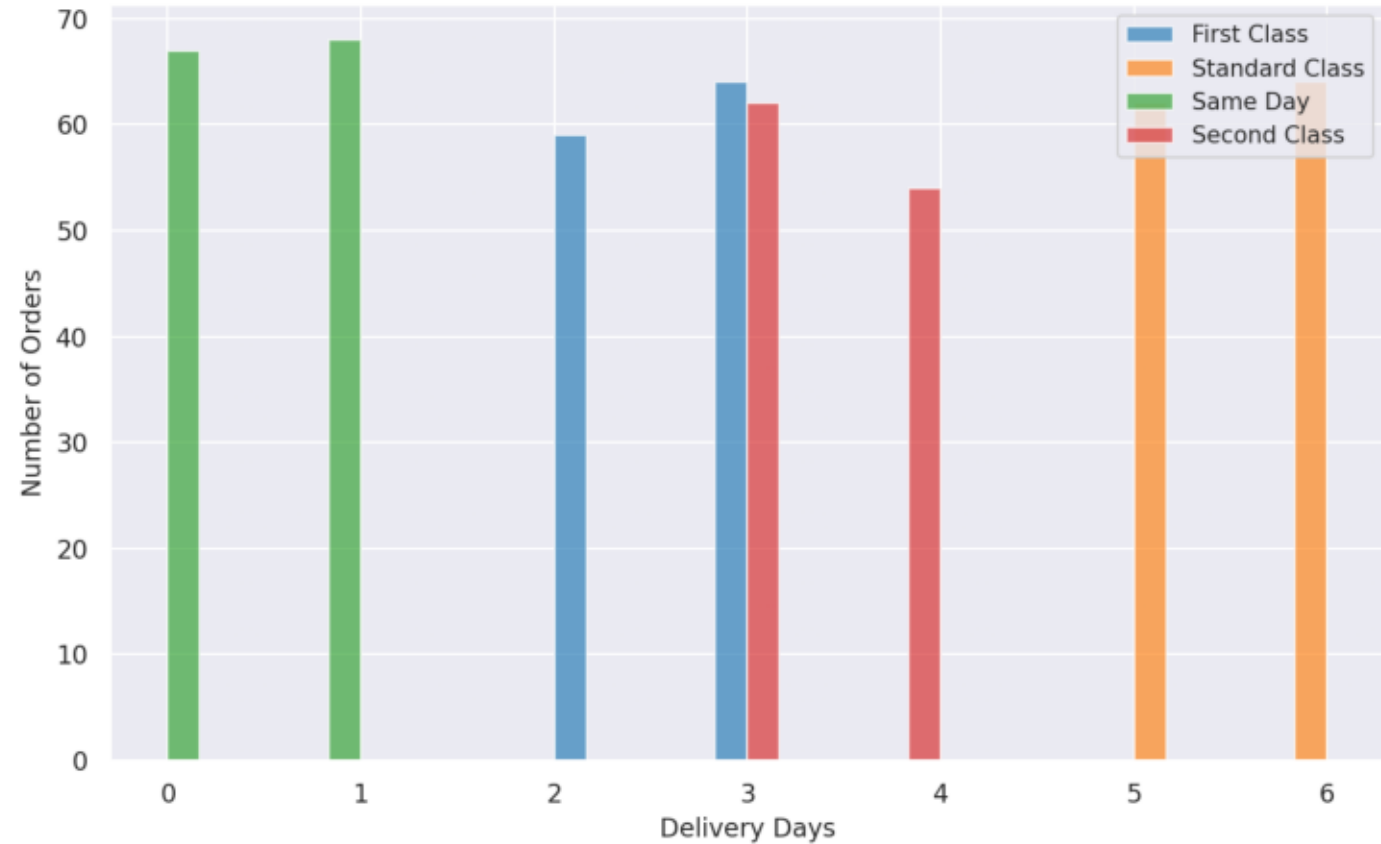


Fig 8: Top 10 Cities by Sales

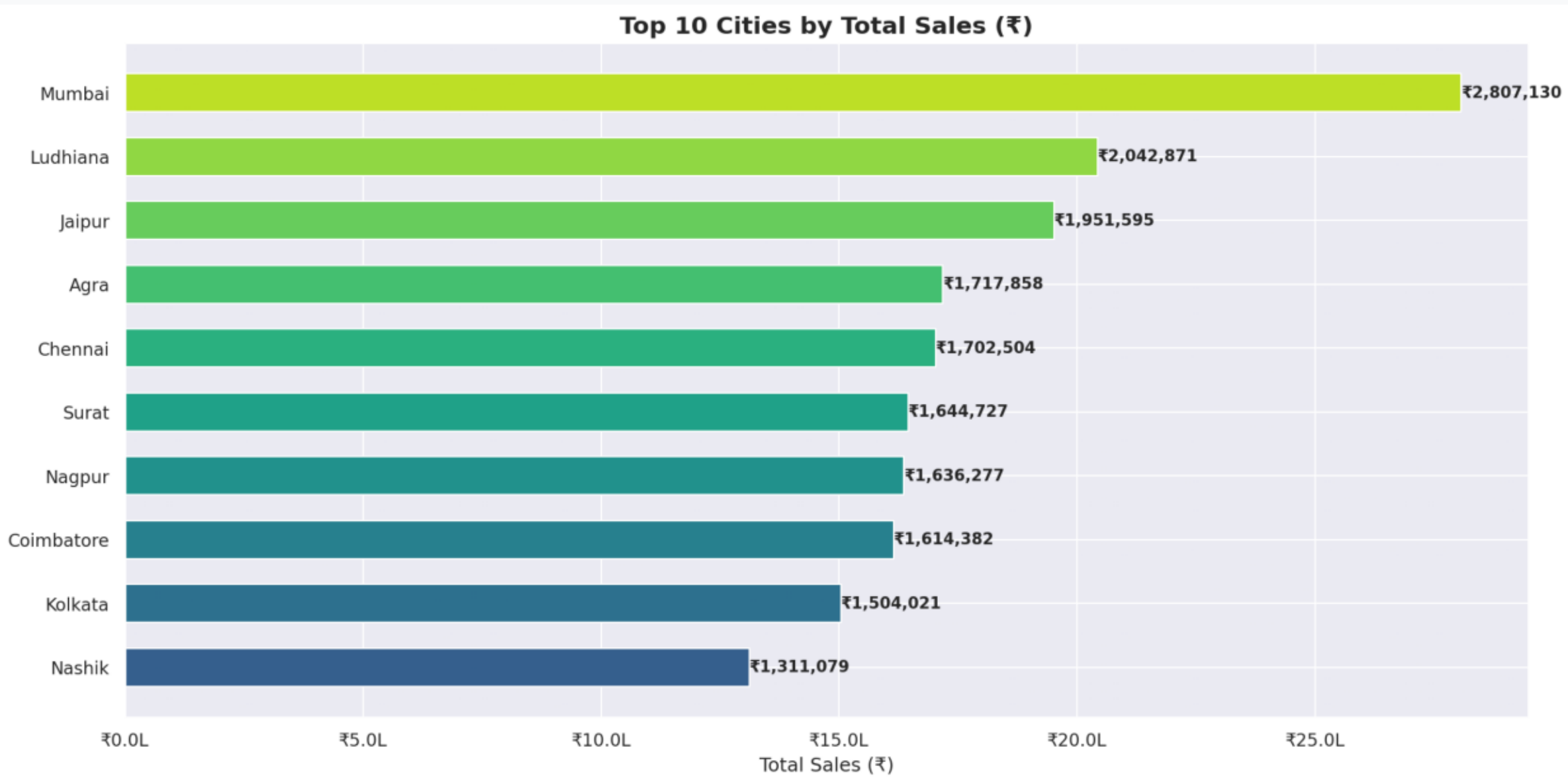


Fig 9: Discount vs Profit Margin

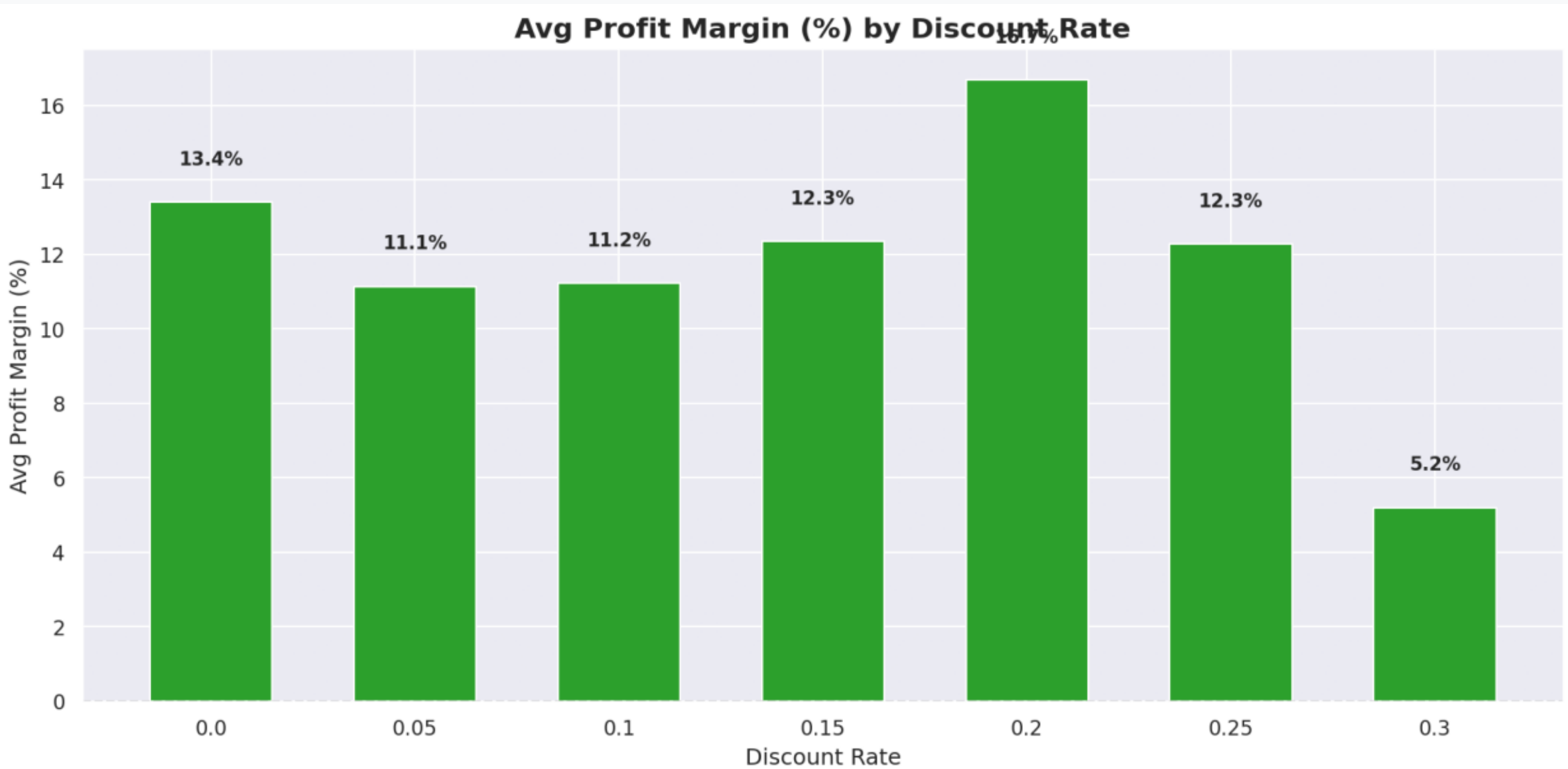
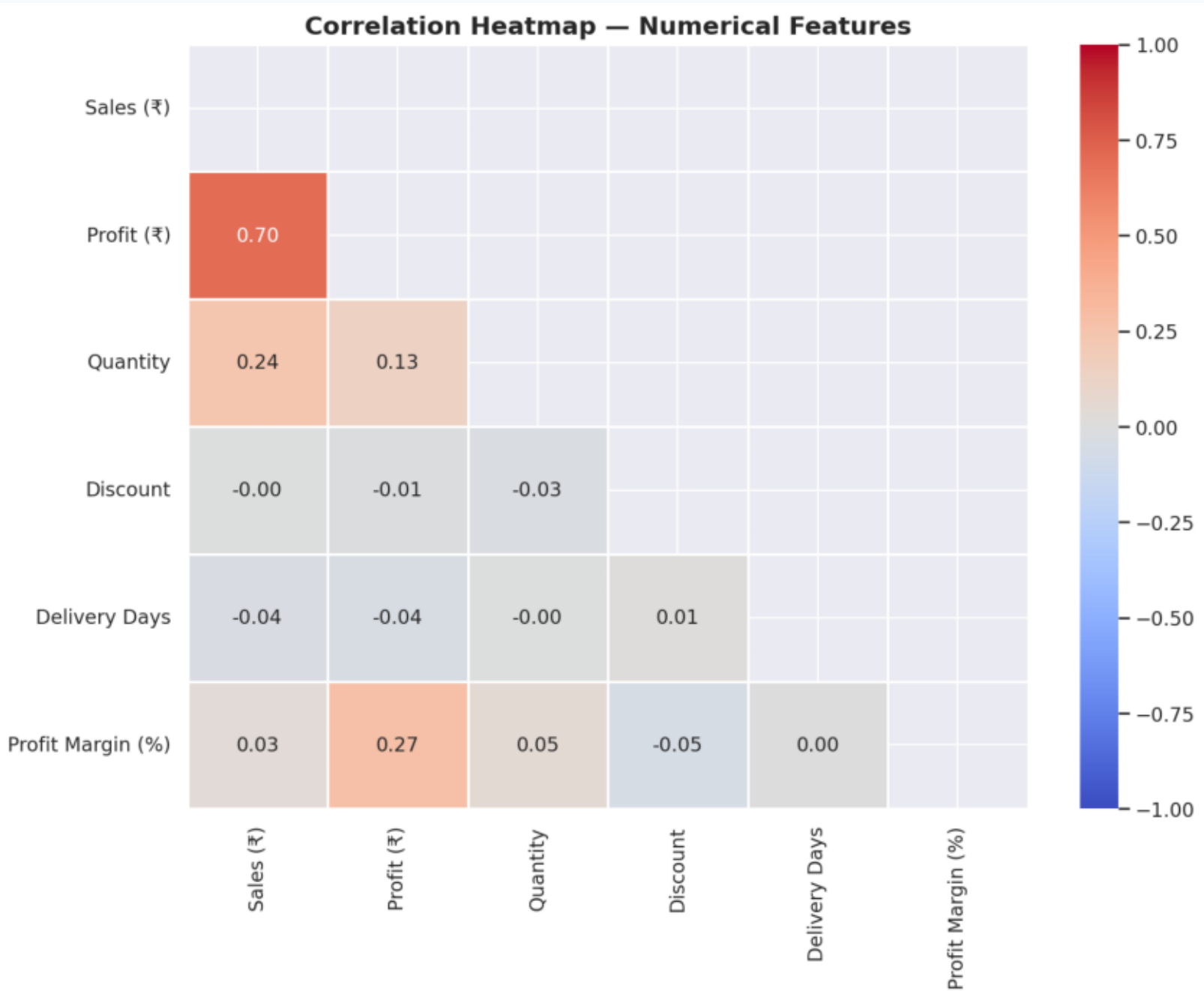


Fig 10: Correlation Heatmap



Data Cleaning Summary Report

Step	Action	Details	Result
1	Load Dataset	515 rows, 18 cols, mixed date formats	Raw CSV created
2	Missing Values	Sales ₹, Profit ₹, Discount → median fill	124 nulls handled
		Customer Name, City → 'Unknown'	
		Postal Code → 0 (integer default)	
3	Drop Duplicates	Exact row matching across all columns	15 rows removed
4	Parse Dates	4 mixed formats → uniform datetime64	2 date columns fixed
5	Type Conversion	Sales/Profit → float2, Qty → Int64	6 columns retyped
6	Feature Engineering	Delivery Days, Month, Year, Profit Margin	4 new columns added
7	Export	global_superstore_cleaned.csv	500 rows × 22 cols